

Prateek Arora

Postdoctoral Researcher, University of California, Berkeley

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Professional Summary

Prateek Arora holds a Ph.D. in Civil Engineering and specializes in structural engineering, probabilistic hazard modeling, and disaster risk analysis. Prateek is committed to advancing methods for assessing natural hazard risks, informing resilient strategies for critical infrastructure, and supporting effective emergency responses. Prateek brings experience in research, teaching, consulting, and cross-disciplinary collaboration to enhance infrastructure resilience to extreme events.

Academic Appointments

University of California, Berkeley, CA

August 2025 - Present

Postdoctoral Researcher, Disaster Risk Analysis Lab

Advisor: Dr. Luis Ceferino

Education

New York University, NY, Ph.D. in Civil Engineering, GPA 4/4

January 2021 - August 2025

Thesis: Probabilistic Modeling for Power Outages and Adoption of Renewables for Enhancing Power Infrastructure Resilience to Extreme Events

Advisor: Dr. Luis Ceferino (now at UC Berkeley), Committee: Dr. Chen Feng, Dr. Semiha Ergan, Dr. Gregory Dobler

NYU Urban Doctoral Fellow, NYU School of Engineering Fellow

Stanford University, CA, M.S. in Structural Engineering, GPA 3.99/4

September 2018 - March 2020

K.C. Mahindra scholarship for graduate studies

BITS Pilani, Pilani Campus, India, B.E. (Hons.) in Civil Engineering, GPA 9.54/10

August 2014 - June 2018

Ranked 1st in the Civil Engineering class of 90 students

Honors, Awards, and Achievements

Best Paper Award, ASCE EMI Objective Resilience Student Competition

2025

Recognized for research leveraging AI and remote sensing for power infrastructure resilience.

Coalition of Disaster Resilient Infrastructure Fellowship, \$15,000

2022

Resilience of Power Grid with Renewables to Hurricanes in a Changing Climate.

New York University Tandon School of Engineering Fellowship

2021

Doctoral Studies in the field of Disaster Risk and Resilience of Urban Environments.

New York University Urban Doctoral Fellowship, \$3,000

2021

Fellowship awarded to 12 Doctoral Students across NYU working towards the development of Urban Cities.

K.C. Mahindra Fellowship, \$2,000

2018

Awarded to 82 out of 1200 applicants from India to pursue graduate studies in the USA.

O.P. Jindal Engineering and Management Scholarship

2016

Awarded for academic and leadership excellence.

Merit Scholarship, BITS Pilani

2017-18

Awarded to the top 3% of students across all disciplines.

National Standard Examinations in Chemistry (INChO)

2013

Ranked among the top 300 students in India eligible for the Indian National Chemistry Olympiad.

National Standard Examinations in Physics (IAPT)

2013

Ranked among the top 1% of students in Rajasthan State in India.

Research Experience

University of California, Berkeley

August 2025 - Present

Postdoctoral Researcher, Disaster Risk Analysis Lab, Advisor: Dr. Luis Ceferino

Berkeley, CA

- Developing optimization frameworks for post-earthquake retrofits using digital twins with MCMC Bayesian updating.
- Collaborating with Pacific Gas and Electric Company for power outage risk models during earthquakes and designing resilience strategies for critical infrastructure.

New York University

Graduate Research and Teaching Assistant

Jan 2021 - August 2025

New York, NY

- Developed probabilistic and machine learning methods for quantifying hurricane-induced power outages.
- Conducted research on the resilience and economic benefits of the adoption of renewables during a hurricane emergency.
- Selected as a Coalition of Disaster Resilient Infrastructure Fellow; One of 15 research proposals selected from 160 submissions.
- Published first-author papers in peer-reviewed journals, presented at international conferences, and delivered invited talks.
- Taught a graduate class of 13 students and an undergraduate class of 16 students.

CSIR-Structural Engineering Research Centre

Research Intern, Structural Health Monitoring Laboratory

July 2017 - December 2017

Chennai, India

- Applied robust statistical methods to account for environmental variability in damage localization.
- Published research article in a peer-reviewed journal on damage identification in bridge structures with limited sensors.

Professional Experience

Rivera Consulting Group, Inc.

Staff Engineer

May 2020 - December 2020

San Francisco, CA

- Designed structural connections and coordinated with architects and building engineers for design requirements.
- Conducted analyses for the retrofit of existing steel-frame, unreinforced masonry, and wood-frame buildings.

Thornton Tomasetti, Inc.

Structural Engineering Intern

June 2019 - September 2019

San Francisco, CA

- Prepared reports for gravity and lateral loads for high-rise buildings according to building codes.
- Reviewed structural shop drawing submittals from contractors to ensure compliance with structural requirements.

Mormugao Port Trust

Summer Intern, Civil Engineering Department

May 2016 - July 2016

Goa, India

- Performed costing and estimation for a railway signaling building at Mormugao Port.
- Prepared construction reports documenting formwork placement and reinforcement detailing.

Working and Submitted Papers

[W3.] **Arora, P.**, Li, C., and Ceferino, L.: Agent-Based Modeling of Power Recovery after Earthquakes in California, [Initial Draft], 2026

[W2.] **Arora, P.**, Ceferino, L., Wang, Z., and Lindsey, M.: Bayesian Updating of Pole-Cable Systems for Damage Quantification after Sequential Extreme Events, [Submitted], 2026

[W1.] Pachev, B., **Arora, P.**, Zhao, J., and Valseth, E.: Global Location-Invariant Peak Storm Surge Prediction, [Under Review], <https://doi.org/10.48550/arXiv.2603.25978>, 2026

Journal Publications

[J7.] **Arora, P.** and Ceferino, L.: Feasibility of Energy Communities for Urban Resilience against Hurricanes, *Natural Hazards Review*, <https://doi.org/10.1061/NHREFO.NHENG-2728>, 2026

[J6.] **Arora, P.**, Ceferino, L. and Dobler, G.: Satellite based inference for disaster induced power outages using daily nighttime light observations, *npj Natural Hazards*, <https://doi.org/10.1038/s44304-026-00270-z>, 2026

[J5.] **Arora, P.** and Ceferino, L.: A Quasi-Binomial Regression Model for Hurricane-Induced Power Outages during Early Warning, *ASCE-ASME Journal of Risk and Uncertainty Analysis in Engineering Systems*, <https://doi.org/10.1061/AJRU66.RUENG-1215>, 2024

[J4.] Lakshmi, K., **Arora, P.** and Rama Mohan Rao, A.: A Global-Local Damage Diagnostic Approach for Large Civil Structures with Very Limited Sensors, *International Journal of Structural Stability and Dynamics*, <https://doi.org/10.1142/S0219455424502705>, 2024

[J3.] Pachev, B., **Arora, P.**, del-Castillo-Negrete, C., Valseth, E., and Dawson, C.: A Framework for Flexible Peak Storm Surge Prediction, *Coastal Engineering*, <https://doi.org/10.1016/j.coastaleng.2023.104406>, 2023

[J2.] **Arora, P.** and Ceferino, L.: Probabilistic and Machine Learning Methods for Uncertainty Quantification in Power Outage Prediction due to Extreme Events, *Natural Hazards and Earth System Sciences*, 23, 1665–1683, <https://doi.org/10.5194/nhess-23-1665-2023>, 2023

[J1.] Singh, M., Yadav, S., Hussain, S., **Arora, P.** and Srivastava, A.: Reliability-Based Mix Design of Marble Dust Incorporated Concrete and Its Assessment Using the Concept of Performance Index, *Structural Concrete*, <https://doi.org/10.1002/suco.202100501>, 2022

Conference Papers and Presentations

[C6.] **Arora, P.***, Cossio, L., Ceferino, L., Wang, Z., and Lindsey, M.: A Digital Twin Approach for Dynamic Risk Assessment Under Sequential Earthquake Hazards, *NHERI Computational Symposium*, Berkeley, CA, USA, 2026.

[C5.] Pachev, B., **Arora, P.***, Zhao, J., and Valseth, E.: Global Peak Storm Surge Prediction, *NHERI Computational Symposium*, Berkeley, CA, USA, 2026.

[C4.] **Arora, P.***, and Ceferino, L.: A Novel Framework for Detecting Disaster-Induced Power Outages Globally via Satellite Observations and Physics-Guided AI, *ASCE Engineering Mechanics Institute (EMI) Conference*, Anaheim, CA, USA, 2025.

[C3.] **Arora, P.***, and Ceferino, L.: A Performance-Based Probabilistic Framework to Model Risk to Power Systems from Hurricanes, *14th International Conference on Applications of Statistics and Probability in Civil Engineering (ICASP14)*, Dublin, Ireland, 2023.

[C2.] **Arora, P.***, and Ceferino, L.: Could Rooftop Solar Panels and Storage Have Enhanced the Electricity Resilience During Hurricane Isaias (2020)?, *14th International Conference on Applications of Statistics and Probability in Civil Engineering (ICASP14)*, Dublin, Ireland, 2023.

[C1.] Athira, G., Sanket, R., Aniruddha, T., **Arora, P.**, and Bahurudeen, A.*: Assessment of Sustainable Alternative Cementitious Materials Using Characterisation Techniques, *Proceedings of Civil Engineering Conference – Innovation for Sustainability (CEC-2016)*, NIT Hamirpur, India, Sept. 9–10, 2016, pp. 430–445.

*Presenter

Technical Reports

[T3.] Wolohan, K., Mosalam, K., Jana, A., Bektas, N., Erazo, K., Garcia, S., Kyprioti, A., Wang, C., Romão, X., Petreski, B., Xu, S., Diekmann, A., Dang, H., **Arora, P.**, Lahna, T. (2024). "StEER: 2024 Hualien City Earthquake Annotated Media Repository", in *StEER – 2024 Hualien City, Taiwan Earthquake*. DesignSafe-CI. <https://doi.org/10.17603/ds2-4xv5-qc41>

[T2.] Cortes, M., **Arora, P.**, Ceferino, L., Ibrahim, H., Istrati, D., Reed, D., Roueche, D., Safiey, A., Tomiczek, T., Zisis, I., Alam, M., Kijewski-Correa, T., Prevatt, D., Robertson, I. (2022). "StEER: Hurricane Ian Preliminary Virtual Reconnaissance Report (PVRr)", in *StEER – 2022 Hurricane Ian*. DesignSafe-CI. <https://doi.org/10.17603/ds2-kc9k-s242>

[T1.] Alam, M., Rodgers, J., Pilkington, S., **Arora, P.**, Gurley, K., Robertson, I. (2022). "StEER: Typhoon Merbok Preliminary Virtual Reconnaissance Report (PVRr)", in *StEER – 17 September 2022 Typhoon Merbok*. DesignSafe-CI. <https://doi.org/10.17603/ds2-4az4-v122>

Open-Source Data and Computational Resources

[O3.] **Arora, P.**, Ceferino, L. (2025). "Power Outage Prediction Utilizing Satellite-based Nightlights Decay," in *Power Outage Prediction Utilizing Satellite-based Nightlights Decay [Version 2]*. DesignSafe-CI. <https://doi.org/10.17603/ds2-a8cv-yc65>

[O2.] **Arora, P.**, Ceferino, L. (2024). "Hurricane Risk Analysis of Power Failures," in *Hurricane Risk Analysis of Power Systems [Version 4]*. DesignSafe-CI. <https://doi.org/10.17603/ds2-a510-6f83>

[O1.] Pachev, B., Del-Castillo-Negrete, C., **Arora, P.**, Dawson, C., Valseth, E. (2023). "Reduced Order Models for Peak Storm Surge in Texas and Alaska," in *ADCIRC Reduced Order Modeling*. DesignSafe-CI. <https://doi.org/10.17603/ds2-85j6-k851>

Workshops and Summer Schools Participation

University of Washington NSF-Funded RAPID Facility Workshop, July 2025. Participated in the NSF RAPID Workshop focused on advancing field reconnaissance, data collection, and resilience modeling for natural hazard response.

Extremes Workshop, Colorado School of Mines (NSF-Funded), November 2024. Attended the interdisciplinary workshop on modeling and analysis of extreme events; engaged in discussions on uncertainty quantification, resilience assessment, and multi-hazard frameworks.

Uncertainty Quantification Summer School, University of Southern California, August 2024. Completed intensive training on uncertainty modeling, Bayesian inference, and stochastic methods for civil and environmental systems.

DesignSafe Hackathon, Texas Advanced Computing Center (NHERI), July 2022. Developed a neural network-based storm surge forecasting model; collaborated with UT Austin researchers to publish a journal article contributing to the field of natural hazard prediction.

North American Regional Science Council Summer School, July 2022. Selected as one of 12 participants for the program on climate change adaptation, engaged in planning exercises to enhance New York City's resilience to storm and flood events.

Talks and Other Participation

Arora, P. (2024). *Power Outage Modeling and Adoption of Renewables Toward Power Resilience During Extreme Weather Events*. Lightning Talk and Poster presented at the Extremes Workshop, Colorado School of Mines, November 2024.

Ceferino, L., & Arora, P. (2023). *Evaluating the Resilience of Hospitals and Power Systems to Earthquakes and Hurricanes*. Invited Talk presented at Thornton Tomasetti, New York, NY, 2023.

Invited participant at the culmination of the fellowship project, Sixth International Conference on Disaster Resilient Infrastructure (ICDRI), 2024.

Grant Writing Experience

NSF FIRE: Fire Science Innovations through Research and Education, submitted, 2025

NSF MATH-DT: Scalable Bayesian Online Learning for Digital Twins in Power Network Resilience against Sequential Extreme Events, awarded, 2025

Bakar Fellows Program, UC Berkeley: OutageIQ-Intelligent Systems for Power Infrastructure Risk Management for Climate and Disaster Resilience, submitted, 2025

Pacific Gas & Electric Company: Statistical Modeling of Power Outages in the Humboldt Region, awarded, 2024

New York City Housing Authority: Climate-Induced Power Outage Recurrence Intervals for Resilience Hubs, submitted, 2023

Coalition of Disaster Resilient Infrastructure: Resilience of Power Grid Infrastructure with Renewables to Extreme Events in a Changing Climate, Co-PI, awarded, 2022

Research Projects

Digital Twin for Power System Resilience under Sequential Extreme Events	August 2025 - Present
<i>Postdoctoral Research, Advisors - Dr. Luis Ceferino, Dr. Ziqi Wang, and Dr. Michael Lindsey</i>	<i>UC Berkeley, CA</i>

- Bayesian updating of structural properties of power poles based on sensor measurements.
- Online updating of the digital twin to optimize the dispatch of repair crews.

Power System Recovery Modeling after Earthquakes	August 2025 - Present
<i>Postdoctoral Research, Advisor - Dr. Luis Ceferino</i>	<i>UC Berkeley, CA</i>

- Agent-based model to identify the recovery sequence of power outages, improving recovery time prediction for customers.
- Modeling the interdependencies of critical infrastructure for comprehensive modeling of power outage recovery.

Hazard Analysis and Urban Utilities Management under Extreme Events	January 2021 - August 2025
<i>Ph.D. Project, Advisor - Dr. Luis Ceferino</i>	<i>New York University, NY</i>

- Study on the feasibility of adopting solar panels during hurricanes, involving modeling hurricane hazard, estimation of available solar energy through empirical models, and fragility analysis of power infrastructure.
- Predicted the power outages due to high winds of a hurricane using machine learning methods.
- Utilized remote sensing and AI to develop a power outage model using satellite-based nightlight observations.

Storm Surge Predictions using Neural Networks

Research in Collaboration with UT Austin

August 2022 - March 2026

New York University, NY

- Investigated the importance of hurricane parameters and bathymetry on storm surge predictions.
- Developed a neural network-based storm surge prediction model to significantly reduce computation time compared to high-fidelity models such as ADCIRC.
- Developed a global storm surge prediction model to predict storm-related surge in different basins.

Data fusion of LIDAR and Imagery for sea waves study

Advisor - Professor Anne Kiremidjian

April 2019 - December 2019

Stanford, CA

- Transforming 2D image data to 3D points based on collinearity condition equations.
- Combining sea wave data from LIDAR, RGB camera, and Infrared camera to identify breaking sea waves.

Structural Health Monitoring

Structural Engineering Research Centre, Advisor - Dr. A Rama Mohan Rao

July 2017 - December 2017

Chennai, India

- Applied method for Damage Identification using Statistical Methods, based on residuals for time-series data using Auto-Regressive Moving Average Model and Kolmogorov Statistical Distance.
- Implemented robust methods to account for variations in SHM data due to different environmental conditions, segregating the outliers in the data from different origins.

Teaching Responsibility

Teaching Assistant

Engineering Mechanics, New York University

August 2023 - December 2023

New York, NY

- Led discussion and tutorial sections for a class of 16 students on structural analysis.
- Conducted weekly office hours to assist graduate students with weekly assignments and class projects.

Teaching Assistant

Disaster Risk Analysis and Urban Systems Resilience, New York University

January 2023 - May 2023

New York, NY

- Developed course material for probabilistic hurricane hazard modeling.
- Led discussions and delivered lectures for a class of 13 students on Disaster Risk Analysis and Urban Systems Resilience.
- Conducted weekly office hours to assist graduate students with weekly assignments and class projects.

Graduate Course Assistant

Introduction to Performance-Based Earthquake Engineering, Stanford University

January 2020 - March 2020

Stanford, CA

- Led discussion sections for a class of 26 students on probabilistic seismic hazard analysis.
- Conducted weekly office hours to assist graduate students with weekly assignments and class projects.

Tutorial Section Leader

Construction Planning and Technology, BITS Pilani

February 2017 - April 2017

Pilani, India

- Delivered a lecture to junior class on designing in Revit Architecture for the Construction Planning and Technology course.

Journal Reviewer

- ASCE-ASME Journal of Risk and Uncertainty Analysis in Engineering Systems
- Journal of Structural Safety and System Reliability
- Frontiers in Energy Research
- ASCE Natural Hazards Review
- International Journal of Disaster Risk Reduction

Mentorship

Graduate Mentor for Capstone Project, University of California, Berkeley January 2025 - May 2025

- Mentored Anna-maria Geofrey Assey and Yufeng Yan for a capstone project for the Master's in Development Engineering.
- Led a team that leveraged satellite data and machine learning to predict power outages during hurricanes.

Summer Undergraduate Research Program, New York University June 2023 - August 2023

- Mentored Dina Hertog-Raz and Tianyi Wang to study solar energy vulnerability to hurricane winds.
- Leveraged power-outage data and solar-panel vulnerability modeling to assess the feasibility of solar generation.

Graduate Mentor for Capstone Project, New York University January 2023 - May 2023

- Mentored Akshay Shetty, Shuang Guo, and Henry Cui for a capstone project at the Center for Urban Science and Progress.
- Team successfully leveraged satellite data and machine learning for power outage predictions during hurricanes.

Summer Undergraduate Research Program, New York University June 2022 - August 2022

- Mentored Richard Liu and Yishi Wang to study a power outage prediction tool and the scalability of solar energy.
- Team leveraged data on real-time power outages and global horizontal solar irradiance.

Mentor to Master's Student, New York University January 2022 - May 2022

- Mentored graduate students to understand the risks to the power system from hurricanes.
- Team leveraged urban nighttime images to identify power outages.

Seismic Design Competition, Earthquake Engineering Research Institute October 2021 - May 2022

- Mentored undergraduate students for the seismic design competition at the National Engineering Earthquake Conference.
- Led the team for seismic analysis of a high-rise building using structural software ETABS.

Positions of Leadership

President, Civil Engineering Association, BITS Pilani January 2018 - May 2018

Publicity Coordinator, Civil Student Society, BITS Pilani August 2016 - May 2017

Professional Credentials

FE Civil

Technical Skills

Probabilistic Risk Analysis, Catastrophe Modeling, Earthquake and Hurricane Risk Analysis, Machine Learning (Python, R), Statistical Modeling, Geospatial Analysis, ArcGIS, MATLAB, Structural and Infrastructure Risk Evaluation, HPC and distributed computing (OpenMPI, multi-node workflows)

Media Coverage

- Paper "Satellite-based inference for disaster-induced power outages using daily nighttime light observations" highlighted by the [United Nations Office for Disaster Risk Reduction \(UNDRR\) PreventionWeb](#), September 2026.
- "New model predicts likely power outages from hurricanes more accurately than conventional predictive techniques," at [New York University](#), July 2023.

Volunteer Activities

- Volunteer, National Service Scheme (NSS), taught high-school students from rural areas near Pilani.
- Student Member, Indian Concrete Institute, August 2016 - August 2017.
- Volunteer, Blood Donation Camp and *Junoon*, a two-day sports meet organized for specially-abled participants.

References

1. **Dr. Luis Ceferino**

Assistant Professor, University of California, Berkeley

Email: ceferino@berkeley.edu

2. **Dr. Ziqi Wang**

Assistant Professor, University of California, Berkeley

Email: ziqi.wang@berkeley.edu

3. **Dr. Eirik Valseeth**

Associate Professor, Norwegian University of Life Sciences

Email: eirik.valseth@nmbu.no

4. **Dr. Gregory Dobler**

Associate Professor, University of Delaware

Email: gdobler@udel.edu